

Math 1522 Calculus 2 Review
Fall 2026

1. Formulas to know.

- (a) Derivative and graph of $\sin^{-1}(x)$
- (b) Derivative and graph of $\cos^{-1}(x)$
- (c) Derivative and graph of $\tan^{-1}(x)$
- (d) Derivative of $\sin(x)$
- (e) Derivative of $\cos(x)$
- (f) Derivative of $\tan(x)$
- (g) Derivative of $\sec(x)$
- (h) State the Inverse Function Theorem
- (i) Derivative and graph of b^x (graph for $b > 1$ and $0 < b < 1$)
- (j) Antiderivative of b^x
- (k) Derivative and graph of $\log_b(x)$ ($b > 1$)
- (l) Antiderivative of x^p if $p \neq -1$ or if $p = -1$
- (m) Definition of $\sinh(x)$
- (n) Definition of $\cosh(x)$

2. Logarithmic differentiation.

(a) Find y' if $y = x^{\sqrt{x}}$

(b) Find y' if $y = \frac{x^{3/4}\sqrt{x^2+1}}{(x^3+x)^3}$

(c) Find y' if $y = (\sin x)^{\sin x}$

(d) Find $y'(0)$ if $y = \frac{(x+1)^4 e^x}{\sqrt{1+3x}}$

3. Derivatives of logarithms

- (a) Find y' if $y = \tan(\ln(x))$
- (b) Find y' if $y = \ln(\sec(x))$
- (c) Find y if $y' = \frac{\sin(x)}{\cos(x)}$
- (d) Find the domain and range of the function $f(x) = \ln(\sqrt{9 - e^x})$
- (e) Explain why we need absolute values when we take the antiderivative of $\frac{1}{x}$.
 - i. Consider the particular antiderivative $\ln|x|$ of $\frac{1}{x}$. Write it as a piecewise function and take its derivative.
 - ii. Compare the domain of this function with the domain of $\frac{1}{x}$
 - iii. Graph both functions on the same set of axes to see that the antiderivative we assign to $\frac{1}{x}$ is the correct most general antiderivative.
- (f) Let $f(x) = 3 - e^{-x}$. Find the domain and range of f , find $f^{-1}(x)$, and state the domain and range of f^{-1} . Does the graph of f have a horizontal asymptote?
- (g) Find $y'(1)$ if $y = 3^x \log_3(x)$.

4. Inverse trig functions

- (a) Prove the identity: $\frac{d}{dx} \sec^{-1}(x) = \frac{1}{x\sqrt{x^2-1}}$ (You may ignore the domain issues with $\sec x$)
- (b) Similar to (a), be able to derive the other derivatives of inverse trig functions.
- (c) Find y' if $y = \cos^{-1}(\sin(x))$
- (d) Find y if $y' = \frac{x^2}{\frac{1}{9}+x^6}$
- (e) Find y if $y' = \frac{1}{\sqrt{a^2-x^2}}$, where $a > 0$. Express y as a function of $\sin^{-1}(x/a)$ or as a function of $\cos^{-1}(x/a)$
- (f) Find the equation of the tangent line to $y = 2 \tan^{-1}(x^2)$ at $x = 1$.
- (g) Write $\cos(\tan^{-1}(x/2))$ as an algebraic function of x (no trig functions).
- (h) Find y if $y' = \frac{x}{\sqrt{1-x^4}}$

5. Hyperbolic functions

- (a) Prove that $\cosh^2 x - \sinh^2 x = 1$
- (b) Find $\cosh^{-1}(2)$ (The equation $\cosh y = 2$ has 2 solutions y , so you should be able to set up some sort of quadratic equation in e^y . Which solution is $\cosh^{-1}(2)$?)
- (c) Prove that the derivative of $\cosh x$ is $\sinh x$ and that the derivative of $\sinh x$ is $\cosh x$
- (d) Evaluate the limit $\lim_{x \rightarrow \infty} \coth x$. Clearly show all algebraic manipulations.
- (e) Find the derivative of $\ln(\cosh(x))$
- (f) Prove that the derivative of $\tanh x$ is $\operatorname{sech}^2 x$
- (g) Find y' if $y = \tan^{-1}(\sinh x)$, and simplify.
- (h) Evaluate $\int_{-\ln 3}^{\ln 3} \cosh x \, dx$

6. Inverse Function Theorem:

- (a) Show the function is 1-1 then find the derivative of the inverse function at the point:
 $f(x) = x + x^3 + 2x^5$, $(4, f^{-1}(4))$
- (b) $f(x) = 5x + 2\sin(x) + 2\cos(x)$, $(2, f^{-1}(2))$
- (c) Explain what could go wrong with using the inverse function theorem on a function that's not 1-1. Draw a picture using $f(x) = x^2$ to illustrate.
- (d) True/False: If a function is 1-1 then it is invertible
- (e) True/False: If a function is invertible then it is 1-1
- (f) True/False: If a function is not continuous then it cannot be 1-1
- (g) True/False: If an invertible function is increasing then its inverse is decreasing
- (h) True/False: If a function is invertible and continuous on an interval then its inverse might be discontinuous
- (i) Let $f(x) = x + e^x$. Find the equation of the tangent line to the graph of f^{-1} at $x = 1$.
- (j) Let $f(x) = 2x^3 + \ln x + \sin(\pi x)$ for $x > 0$. You may assume f is 1-1. Find the slope of the tangent line to the graph of f^{-1} at $x = 2$. (Hint: what is $f(1)$?)

7. L'Hôpital's rule.

(a) $\lim_{x \rightarrow \infty} x e^{-x}$

(b) $\lim_{x \rightarrow 0^+} x \ln x$

(c) $\lim_{x \rightarrow 0^+} x^x$

(d) $\lim_{x \rightarrow 0^+} x^{\frac{1}{x}}$

(e) $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

(f) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$

(g) $\lim_{x \rightarrow 0^+} (1 + \sin(2x))^{\cot x}$

(h) $\lim_{x \rightarrow \infty} (e^x - 2^x)$

(i) $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{e^x - 1}\right)$

(j) $\lim_{x \rightarrow 1^-} \frac{\ln x}{\cos^{-1}(x)}$

(k) $\lim_{x \rightarrow \infty} x \left(\frac{\pi}{2} - \tan^{-1}(2x)\right)$

(l) $\lim_{x \rightarrow \infty} \sin^{-1} \left(\frac{e^x}{2e^x + 1}\right)$

(m) $\lim_{x \rightarrow \infty} \frac{\cosh(\ln x)}{x}$

8. Substitution. Fully evaluate the antiderivative, then plug in the bounds. Or change the bounds with your substitution to evaluate while in terms of u . Clearly declare each substitution.

(a) $\int_0^{(\frac{\pi}{2})^{\frac{1}{3}}} x^2 \sin x^3 dx$
(b) $\int_0^1 (2x+2)e^{x^2+2x+3} dx$
(c) $\int_0^1 \frac{e^{\tan^{-1}(x)}}{1+x^2} dx$
(d) $\int_{\pi/4}^{\pi/2} \cot x dx$
(e) $\int_1^e \frac{(\ln x)^2}{x} dx$
(f) $\int \frac{\cos x}{4+\sin^2 x} dx$
(g) $\int \frac{dx}{\sqrt{x}(1+\sqrt{x})}$

9. Integration by parts. Fully evaluate the antiderivative, then after, plug in the bounds. Clearly declare each substitution.

(a) $\int_0^1 x^2 e^x dx$
(b) $\int_1^e x^4 \ln(x) dx$
(c) $\int_1^e \ln(x) dx$
(d) $\int_0^1 \tan^{-1}(x) dx$
(e) $\int_0^\pi e^x \sin(x) dx$
(f) $\int_0^{\pi/2} \sin(x) \cos(x) dx$
(g) $\int x \sec^2(x) dx$
(h) $\int x \sin(2x) dx$
(i) $\int_0^1 x \tan^{-1}(x) dx$

10. Trig integrals. Find the most general antiderivative.

(a) $\int \sin^3(x) \cos^2(x) dx$
(b) $\int \sin^3(x) dx$
(c) $\int \cos^3(x) \sin^2(x) dx$
(d) $\int \sin^2(x) \cos^2(x) dx$
(e) $\int \cos^2(x) dx$
(f) $\int \tan^2(x) dx$
(g) $\int \tan^3(x) dx$
(h) $\int \tan^2(x) \sec^4(x) dx$
(i) $\int \sin(3x) \cos(2x) dx$
(j) $\int_0^{\pi/3} \tan(x) \sec^3(x) dx$ (evaluate at the bounds)

11. Trig substitution. Find the most general antiderivative then evaluate it at the bounds, or substitute the bounds with each substitution. Clearly declare each substitution.

(a) $\int_2^3 \frac{dx}{(x^2-1)^{\frac{3}{2}}}$

(b) $\int_0^2 \frac{dx}{\sqrt{4+x^2}}$

(c) $\int_0^1 \frac{dx}{\sqrt{1+4x^2}}$

(d) $\int_0^2 \frac{dx}{\sqrt{4+\frac{1}{4}x^2}}$

(e) $\int_0^1 \frac{dx}{\sqrt{4+9x^2}}$

complete the square:

(f) $\int \frac{dx}{\sqrt{x^2+4x-5}}$

(g) $\int \frac{dx}{(6-8x-x^2)^{\frac{3}{2}}}$

more practice:

(h) $\int_0^{\sqrt{3}} \frac{x^3}{\sqrt{x^2+1}} dx$

(i) $\int_{\sqrt{2}}^2 \frac{dx}{x^2\sqrt{x^2-1}}$

(j) $\int_0^1 \frac{dx}{(1+x^2)^2}$

12. Partial fractions

- (a) With long division. Find the most general antiderivative.

i. $\int \frac{x^2}{x-1} dx$

ii. $\int \frac{3x^2+6x+2}{x^2+3x+2} dx$

iii. $\int \frac{x^3+4}{x^2+4} dx$

- (b) With fully factorable denominators. Find the most general antiderivative.

i. $\int \frac{dx}{(x^2-1)^2}$

ii. $\int \frac{2x dx}{2x^2+3x+1}$

iii. $\int \frac{e^{2x} dx}{e^{2x}+3e^x+2}$

iv. $\int \frac{x+4}{x^3-4x} dx$

(c) With irreducible quadratics. Find the most general antiderivative.

i. $\int \frac{2x+5}{(x-2)^2(x^2+2)} dx$

ii. $\int \frac{3x^2-x+8}{x^3+4x} dx$

iii. Set up, but do not solve, the partial fraction decomposition of $\frac{x+1}{x^2(x-3)(x^2+1)^2}$

13. Improper integrals. Set-up and evaluate a limit expression to determine if the integral converges or diverges. If it diverges, describe an infinite area that the graph of the function encloses. If it converges determine what it converges to, unless indicated otherwise.

(a) $\int_0^1 \ln(x) dx$

(b) $\int_0^1 \frac{\ln x}{x} dx$

(c) $\int_1^\infty \frac{dx}{x}$

(d) $\int_1^\infty \frac{dx}{\sqrt{x}}$

(e) $\int_1^\infty \frac{dx}{x^{0.9}}$

(f) $\int_1^\infty \frac{dx}{x^p}$ (consider $p \leq 1$ and $p > 1$)

(g) $\int_0^\pi \tan(x) dx$

(h) $\int_{-1}^1 \frac{dx}{x}$

(i) $\int_{-4}^2 \frac{x-1}{(x+2)^2} dx$

(j) $\int_0^\infty x e^{-x} dx$

(k) $\int_0^\infty \sin(x) dx$

(l) $\int_0^\infty e^{x^2} dx$ (don't determine the value)

(m) $\int_0^\infty 2^{\sin x} dx$ (don't determine the value)

(n) $\int_0^\infty \frac{e^x}{(1+e^x)^3} dx$

(o) $\int_1^\infty \frac{\ln x}{x^3} dx$

(p) $\int_0^2 \frac{dx}{(x-1)^{2/3}}$

(q) $\int_2^\infty \frac{dx}{4+x^2}$

14. Numerical integration. From the function, determine which methods will give an over estimate, and which methods will give an under estimate. The list of methods is: Left endpoint rule, Right endpoint rule, Midpoint rule, Trapezoid rule. After, confirm you were correct for the midpoint and trapezoid rules by calculating the estimated area given by midpoint rule and and trapezoid rule with $n=4$. You may use a calculator.

(a) $\int_0^4 e^x dx$

(b) $\int_0^\pi \sin(x) dx$

(c) $\int_0^1 (x^3 - 3x^2 + 2x + 1) dx$

(d) Use the trapezoid rule with $n = 3$ to approximate $\int_0^3 2^x dx$. Leave your answer in exact form.

- (e) The error in the trapezoid rule satisfies $|E_T| \leq \frac{K(b-a)^3}{12n^2}$, where $|f''(x)| \leq K$ on $[a, b]$. How large must n be to guarantee that the trapezoid rule approximates $\int_1^2 \frac{dx}{x}$ with error less than 10^{-4} ?

15. Sequences.

- Give an example of a sequence that converges to 0.
- Give an example of a sequence that converges to 1
- give an example of a bounded sequence that diverges.
- give an example of a sequence that diverges to infinity
- If an alternating sequence converges, what must it converge to?
- Given a series $\sum_{n=0}^{\infty} a_n$, what are the 2 sequences that are relevant to this series? What are their limits?
- Write the sequence as a function whose domain is the natural numbers (e.g. $\{1, 4, 9, 16, \dots\} = \{f(n)\}_{n=0}^{\infty}$ where $f(n) = n^2$, $f : \mathbb{N} \rightarrow \mathbb{R}$. You may change notation to write $f(n) = a_n$)
 $\{1, \frac{1}{4}, \frac{1}{16}, \frac{1}{64}, \dots\}$
- Write the sequence as a function whose domain is the natural numbers
 $\{\frac{-1}{8}, \frac{1}{27}, \frac{-1}{64}, \frac{1}{125}, \dots\}$
- Write the sequence as a function whose domain is the natural numbers
 $\{1, 1, 2, 6, 24, 120, \dots\}$
- For each sequence, decide if it is increasing, decreasing, or not monotonic, and if it is bounded or unbounded. $a_n = 1 + \frac{(-1)^n}{n}$ ($n \geq 1$) and $b_n = \frac{n}{n+1}$ ($n \geq 1$)
- Show that $\lim_{n \rightarrow \infty} \frac{3^n}{n!} = 0$

16. Test for Divergence (TFD)

- $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$
- $\sum_{m=1}^{\infty} (\frac{4}{3})^m$
- $\sum_{n=0}^{\infty} \frac{n^2+1}{6n^2+6n+1}$
- $\sum_{n=0}^{\infty} \sin(n)$
- $\sum_{n=1}^{\infty} 2^{1/n}$
- Can you use the TFD to determine if $\sum_{n=2}^{\infty} \frac{1}{n^2}$ converges or diverges? Explain why or why not.
- Suppose $b_n > 0$ for all n and $\sum_{n=1}^{\infty} b_n$ converges. What does the TFD say about $\sum_{n=1}^{\infty} \frac{1}{b_n}$?

17. Telescoping series

- Evaluate the series $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$
- Evaluate the series $\sum_{n=1}^{\infty} \frac{2}{4n^2-1}$
- Evaluate the series $\sum_{n=1}^{\infty} [\tan^{-1}(n+2) - \tan^{-1}(n)]$

- (d) Evaluate the series $\sum_{n=1}^{\infty} [f(n) - f(n+1)]$, where $\lim_{n \rightarrow \infty} f(n) = L$
- (e) Evaluate the series $\sum_{n=1}^{\infty} [f(n) - f(n+k)]$, where k is a positive integer and $\lim_{n \rightarrow \infty} f(n) = L$
- (f) Determine if the series converges or diverges by writing out its partial sums $\sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n}\right)$

18. Geometric series

- (a) Evaluate the series $\sum_{n=0}^{\infty} \left(\frac{3}{4}\right)^n$
- (b) Evaluate the series $\sum_{n=0}^{\infty} \frac{4^{2n-1}}{17^n}$
- (c) Evaluate the series $\sum_{n=-4}^{\infty} \left(\frac{1}{2}\right)^n$
- (d) Evaluate the series $\sum_{n=1}^{\infty} \frac{3^{n+1}}{4^{n-1}}$
- (e) Evaluate the series $2 - 1 + \frac{1}{2} - \frac{1}{4} + \frac{1}{8} - \dots$
- (f) For which real numbers a does $\sum_{k=1}^{\infty} \left(\frac{2a}{1+a^2}\right)^k$ converge? Find its value (in terms of a) when it converges.

19. Integral Test

- (a) Determine if the series converges or diverges $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$
- (b) Determine if the series converges or diverges $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)}$
- (c) Show that the series converges if $p > 1$ and diverges if $0 < p < 1$. Also show it diverges if $p = 1$. $\sum_{n=1}^{\infty} \frac{1}{n^p}$
- (d) Determine if the series converges or diverges $\sum_{n=1}^{\infty} ne^{-n^2}$
- (e) Can you use the integral test to determine if $\sum_{n=1}^{\infty} \frac{2+\sin n}{n^2}$ converges or diverges? Explain why or why not. If not, which test works?

20. p-series

- (a) Determine if the series converges or diverges $\sum_{n=5}^{\infty} \frac{1}{n^e}$
- (b) Determine if the series converges or diverges $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n}}$
- (c) Determine if the series converges or diverges $\sum_{n=1}^{\infty} \frac{1}{n^{0.000001}}$
- (d) Determine if the series converges or diverges $\sum_{n=1}^{\infty} \frac{n^2 + \sqrt{n}}{n^4}$
- (e) For which real numbers p does $\sum_{n=1}^{\infty} \frac{1}{n^{2p-1}}$ converge?

21. Direct Comparison Test (DCT) and Limit Comparison Test (LCT)

- (a) Determine if the series converges or diverges $\sum_{n=2}^{\infty} \frac{1}{\ln n}$
- (b) Determine if the series converges or diverges $\sum_{n=1}^{\infty} \sin(1/n)$
- (c) Determine if the series converges or diverges $\sum_{n=0}^{\infty} \frac{\tan^{-1}(n)}{n^2+1}$
- (d) Determine if the series converges or diverges $\sum_{n=0}^{\infty} \frac{n+1}{n^2+n+1}$
- (e) Determine if the series converges or diverges $\sum_{n=1}^{\infty} \frac{3n+2}{n^{1.99}+2}$
- (f) Determine if the series converges or diverges $\sum_{n=1}^{\infty} \frac{1}{2^n+n}$

- (g) Determine if the series converges or diverges $\sum_{n=1}^{\infty} \frac{n^2-2n}{n^{7/2}+1}$
- (h) Determine if the series converges or diverges $\sum_{n=1}^{\infty} \frac{1}{n^2+n \cos n}$

22. Alternating Series Test (AST)

- (a) Determine if the series converges or diverges $\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$
- (b) Determine if the series converges or diverges $\sum_{i=1}^{\infty} \frac{\cos(\pi i)}{i^{0.1}}$
- (c) Determine if the series converges or diverges $\sum_{k=2}^{\infty} \frac{(-1)^{k+15}}{\ln k}$
- (d) Determine if the series converges or diverges $\sum_{n=1}^{\infty} \frac{(-1)^n \ln n}{\sqrt{n}}$
- (e) Determine if the series converges or diverges $\sum_{n=1}^{\infty} \frac{(-1)^n \sqrt{n}}{n+4}$
- (f) Let $a_n = \frac{1}{n}$ if n is odd and $a_n = \frac{1}{n^2}$ if n is even. Can you use the AST to determine if $\sum_{n=1}^{\infty} (-1)^n a_n$ converges? Explain why or why not.

23. Ratio Test

- (a) Determine if the series converges or diverges $\sum_{n=0}^{\infty} \frac{1}{n!}$
- (b) Determine if the series converges or diverges $\sum_{n=0}^{\infty} \frac{n!}{n^n}$
- (c) Determine if the series converges or diverges (for this one you could theoretically use TFD) $\sum_{n=0}^{\infty} \frac{(2n+1)!}{3^n}$
- (d) Determine if the series converges or diverges $\sum_{n=1}^{\infty} \frac{n^2 4^n}{n!}$

24. Root Test

- (a) Determine if the series converges or diverges $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1}\right)^n$
- (b) Determine if the series converges or diverges $\sum_{n=1}^{\infty} \frac{n^n}{2n^2}$
- (c) Show that the Root Test is inconclusive for both $\sum_{n=1}^{\infty} \frac{1}{n}$ and $\sum_{n=1}^{\infty} \frac{1}{n^2}$. Which one converges?
- (d) Consider $\sum_{n=1}^{\infty} \frac{2^n}{n^5}$. Can you use the Root Test to determine if it converges or diverges? Can you use the Ratio Test? Explain.

25. Absolute and conditional convergence

- (a) Determine if the series converges absolutely, converges conditionally, or diverges $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$
- (b) Determine if the series converges absolutely, converges conditionally, or diverges $\sum_{l=1}^{\infty} \frac{(-1)^l}{l^2}$
- (c) Determine if the series converges absolutely, converges conditionally, or diverges $\sum_{k=1}^{\infty} \frac{k \cos \pi k}{k^3+1}$
- (d) Determine if the series converges absolutely, converges conditionally, or diverges $\sum_{n=1}^{\infty} \frac{(-1)^n n^3}{2^n}$
- (e) Determine if the series converges absolutely, converges conditionally, or diverges $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2+1}$

- (f) Suppose $\sum_{n=1}^{\infty} a_n$ converges to 2. What is $\lim_{n \rightarrow \infty} a_n$? Must $\sum_{n=1}^{\infty} |a_n|$ also converge?

26. Radius of Convergence (ROC) and Interval of Convergence (IOC)

- (a) Find the ROC and IOC. $\sum_{n=0}^{\infty} n^n x^n$
- (b) Find the ROC and IOC. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$
- (c) Find the ROC and IOC. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$
- (d) Find the ROC and IOC. $\sum_{n=0}^{\infty} \left(\frac{-3x}{2}\right)^n$
- (e) Find the ROC and IOC. $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{3^n (2n+1)}$
- (f) Find the ROC and IOC. $\sum_{n=1}^{\infty} \frac{(x+2)^n}{n 4^n}$
- (g) Find the center, ROC, and IOC. $\sum_{n=1}^{\infty} \frac{2^n (x-3)^n}{n^2}$

27. Power series from geometric series

- (a) Find the power series that represents the function at the center $x=a$. $f(x) = \frac{1}{1+x}$, $a = 0$. State the ROC and IOC
- (b) Find the power series that represents the function at the center $x=a$. $g(x) = \frac{1}{1+x}$, $a = 1$. State the ROC and IOC.
- (c) Find the power series that represents the function at the center $x=a$. $g(x) = \frac{1}{1+x}$, $a = -3$. State the ROC and IOC.
- (d) Find the power series that represents the function at the center $x=a$. $f(x) = \frac{1}{1-x^3}$, $a = 0$. State the ROC and IOC.
- (e) Find the power series that represents the function at the center $x=a$. $g(x) = \tan^{-1}(x)$, $a = 0$. State the ROC
- (f) Find the power series that represents the function at the center $x=a$. $g(x) = \ln(x)$, $a = 1$. State the ROC
- (g) Find the power series that represents the function at the center $x=a$, $f(x) = \frac{1}{4-x^2}$, $a = 0$. For fun also try to find it by writing $f(x) = \frac{1}{2-x} \frac{1}{2+x}$ and performing a multiplication of power series.
- (h) Find the power series that represents the function at the center $x=a$, $f(x) = \frac{2}{(1+x)^3}$, $a = 0$
- (i) Find the function that the power series represents $\sum_{n=2}^{\infty} 2n(n-1)x^n$
- (j) Find the power series that represents the function at the center $x=a$, $f(x) = \frac{x}{(1-x^2)^2}$, $a = 0$ (Hint: differentiate $\frac{1}{1-x^2}$)
- (k) Find the exact value of $\sum_{n=1}^{\infty} \frac{n}{3^n}$

28. Taylor and Maclaurin series

- (a) Find the Maclaurin Series for e^x , $\sin(x)$, $\cos(x)$
- (b) Find the Maclaurin series for $\cosh(x)$, and $\sinh(x)$
- (c) Find the degree 2 Taylor polynomial of $f(x) = \sqrt{x}$ centered at $a = 9$
- (d) Find the degree 3 Taylor polynomial of $f(x) = \cos(x)$ centered at $a = \frac{\pi}{3}$

- (e) Find the Maclaurin series for $F(x) = \int_0^x e^{-t^2} dt$ and state its ROC
- (f) Find the exact value of $1 - \frac{\pi^2}{3^2 \cdot 2!} + \frac{\pi^4}{3^4 \cdot 4!} - \frac{\pi^6}{3^6 \cdot 6!} + \dots$
- (g) Find the exact value of $\sum_{n=0}^{\infty} \frac{(\ln 3)^n}{n!}$
- (h) Find the exact value of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n 2^n}$
- (i) Use a Maclaurin series to evaluate $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$

29. Approximations and error bounds

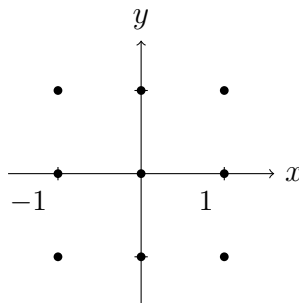
- (a) Approximate the value of the integral to within 10^{-6} . $\int_0^{0.1} \frac{dx}{1+x^4}$
- (b) Use a degree 3 polynomial, $p_3(x)$ to approximate $\tan^{-1}(x)$ near $x=0$. Use Alternating Series Estimation to find an upperbound for the error in using $p_3(1/2)$ to approximate $\tan^{-1}(1/2)$
- (c) Find an upper bound for the same error using Taylor's Inequality. To get M , bound $|f^{(4)}(x)|$ on the interval $[0, 1/2]$, where $f(x) = \tan^{-1}(x)$.
- (d)
 - i. Find the function that represents the power series $f(x) = \sum_{n=2}^{\infty} 5n(n-1)x^n$
 - ii. Use that function to evaluate $f(1/2) = \sum_{n=2}^{\infty} 5n(n-1)(\frac{1}{2})^n$.
 - iii. Find $p_2(x)$, the quadratic approximation for f centered at 0.
 - iv. Find $p_2(1/2)$.
 - v. Calculate the exact error $|f(1/2) - p_2(1/2)|$
 - vi. calculate the upper bound on the error given by Taylor's inequality for $x=1/2$, with M the maximum of $|f'''(x)|$ on $[0, 1/2]$. You can use the 3rd derivative of $f(x)$ is $\frac{600x^2}{(1-x)^6} + \frac{720x}{(1-x)^5} + \frac{180}{(1-x)^4}$
- (e) Approximate $\ln(1.1)$ to within 0.01 (Use ASE)
- (f) Approximate $\ln(0.9)$ to within 0.01 (Use Taylor's Inequality)
- (g) Approximate $\sin(0.1)$ to within 10^{-5}
- (h) How many terms of $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ are needed to approximate its sum to within 0.001?
- (i) Let $f(x) = e^x$. Find $T_3(x)$, the degree 3 Taylor polynomial of f centered at 0, and use Taylor's Inequality to bound $|f(x) - T_3(x)|$ for $-1 \leq x \leq 1$.
- (j) Approximate $\int_0^1 e^{-x^2} dx$ to within 0.01 (Use ASE)

30. Complex numbers

- (a) Compute $(1+i)^{10}$ write your answer in $a+bi$ form
- (b) State Euler's Formula
- (c) Compute $(i-1)^{12}$ write your answer in $a+bi$ form
- (d) convert $3+4i$ to polar coordinates
- (e) convert $2e^{i\pi/4}$ to rectangular coordinates

31. Differential equations

- (a) Show that $y = Ce^{x^2}$ is a solution of $y' = 2xy$ for every constant C . Then find the solution with $y(0) = 3$.
- (b) Sketch the direction field of $y' = x + y$ at the nine points (x, y) with $x, y \in \{-1, 0, 1\}$. Along which line are all the slopes 0? Then sketch the solution curve through $(0, 0)$.



- (c) Find the equilibrium solutions of $y' = y^2 - 4$. For each one, use the sign of y' to decide if nearby solutions move toward it or away from it.
- (d) Use Euler's method with step size $h = 0.5$ to approximate $y(1)$, where $y' = x + y$ and $y(0) = 1$. Is your approximation an over estimate or an under estimate? (Hint: what is y'' ?)
- (e) Use Euler's method with step size $h = -1$ to approximate $y(-1)$, where $y' = y - x$ and $y(1) = 3$.
- (f) Solve the initial value problem $\frac{dy}{dx} = xy^2$, $y(0) = 1$. On what interval is the solution defined?
- (g) Solve the initial value problem $\frac{dy}{dx} = e^{x-y}$, $y(0) = \ln 2$.
- (h) Solve the initial value problem $\frac{dy}{dt} = 3y - 6$, $y(0) = 5$.
- (i) A population of bacteria grows at a rate proportional to its size. It starts with 200 cells and has 600 cells after 2 hours. Find the population $P(t)$ after t hours, and find when the population is 5400.
- (j) A population satisfies the logistic equation $\frac{dP}{dt} = 0.5P \left(1 - \frac{P}{1000}\right)$ with $P(0) = 200$.
- What is the carrying capacity? What are the equilibrium solutions?
 - For which values of P is the population increasing? For which value of P is it increasing fastest?
 - Solve for $P(t)$, and find when $P = 500$.